

Zebra FX9600 Fixed UHF RFID Reader

The Zebra FX9600 is a high-performance fixed UHF RFID reader designed for high-volume, demanding environments. With up to 8 highly sensitive monostatic RF ports, IP53 die-cast aluminum housing, support for PoE/PoE+, and on-reader integrated applications, it delivers industry-leading RFID performance for warehouse, loading dock, and supply chain logistics.



PHYSICAL CHARACTERISTICS

Dimensions	27.3 cm L x 18.4 cm W x 5 cm D
Weight	Approx. 4.4 lb / 2.13 kg
Housing Material	Die-cast aluminum, IP53
Visual Status Indicators	Multicolor LEDs: Power, Activity, Status and Applications

RFID SPECIFICATIONS

Max Receive Sensitivity	-86 dBm monostatic
Air Protocol	ISO 18000-63 (EPC Class 1 Gen 2 V2)
Frequency (UHF Band)	Global: 902-928 MHz (and partial-band countries), 865-868 MHz; USA-only: 902-928 MHz
Transmit Power Output	0 dBm to +33.0 dBm (PoE+, 24V Ext. DC, Universal 24 VDC PSU); 0 dBm to +31.5 dBm (PoE, 12V Ext. DC for 4-port, 24V Ext. DC, Universal 24 VDC PSU)

CONNECTIVITY

Connections	10/100 BaseT Ethernet (RJ45); USB Host & Client (Type A & B); Serial (DB9)
General Purpose I/O	4 in, 4 out, optically isolated (terminal block)
Power	PoE (802.3af); PoE+ (802.3at); +24V DC (UL approved)
Antenna Ports	FX9600-4: 4 monostatic ports (RP-TNC); FX9600-8: 8 monostatic ports (RP-TNC)

ENVIRONMENTAL

Operating Temperature	-4 °F to +131 °F / -20 °C to +55 °C
Storage Temperature	-40 °F to +158 °F / -40 °C to +70 °C
Humidity	5% to 95% non-condensing
Housing	IP53

HARDWARE, OS AND FIRMWARE

Processor	Texas Instruments AM3505 (600 MHz)
Memory	Flash 512 MB; DRAM 256 MB
Operating System	Linux
Firmware Upgrade	Web-based and remote firmware update capabilities
Management Protocols	RM 1.0.1 (HTTP / HTTPS via XML and SNMP); RDMP
Network Services	DHCP, HTTPS, FTPS, SFTP, SSH, HTTP, FTP, SNMP, NTP
Network Stack	IPv4 and IPv6
Security	Transport Layer Security v1.2, FIPS-140
API Support	Host applications – .NET, C and Java EMDK; Integrated applications – C, Java SDK

COMPLIANCE

Safety	UL 60950-01, UL 2043, IEC 60950-1, EN 60950-1
RF/EMI/EMC	FCC Part 15, RSS 210, EN 302 208, ICES-003 Class B, EN 301 489-1/3; Malaysia: 919-923 MHz
SAR/MPE	FCC 47CFR2:OET Bulletin 65; EN 50364
Other	RoHS, WEEE

FEATURE HIGHLIGHTS

- High RF sensitivity for superior read range and accuracy
- Durable die-cast aluminum housing with IP53 rating
- Supports PoE and PoE+ for simplified installation
- Up to 8 antenna ports per reader (FX9600-8 model)
- On-reader integrated applications for edge analytics
- Wi-Fi & Bluetooth connectivity via dongle
- Optional GPS-capable 4G LTE cellular modem

PRODUCT GROUP

Category	RFID Reader
Product Group	Fixed UHF RFID Reader